Assembly manual for the electrical of PNPv3

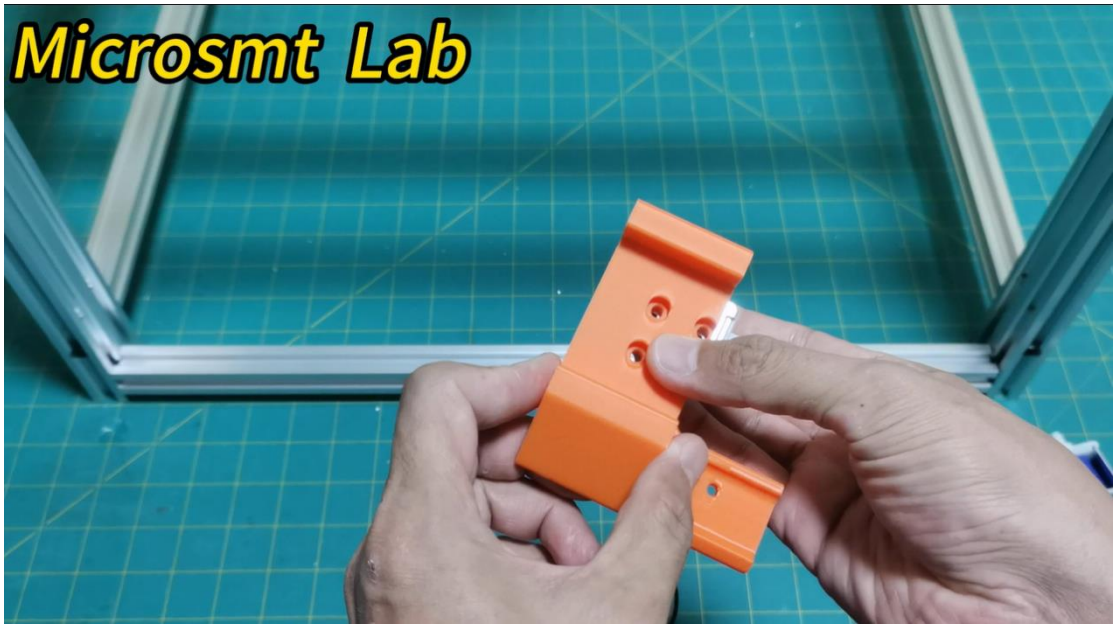
2025.08.28

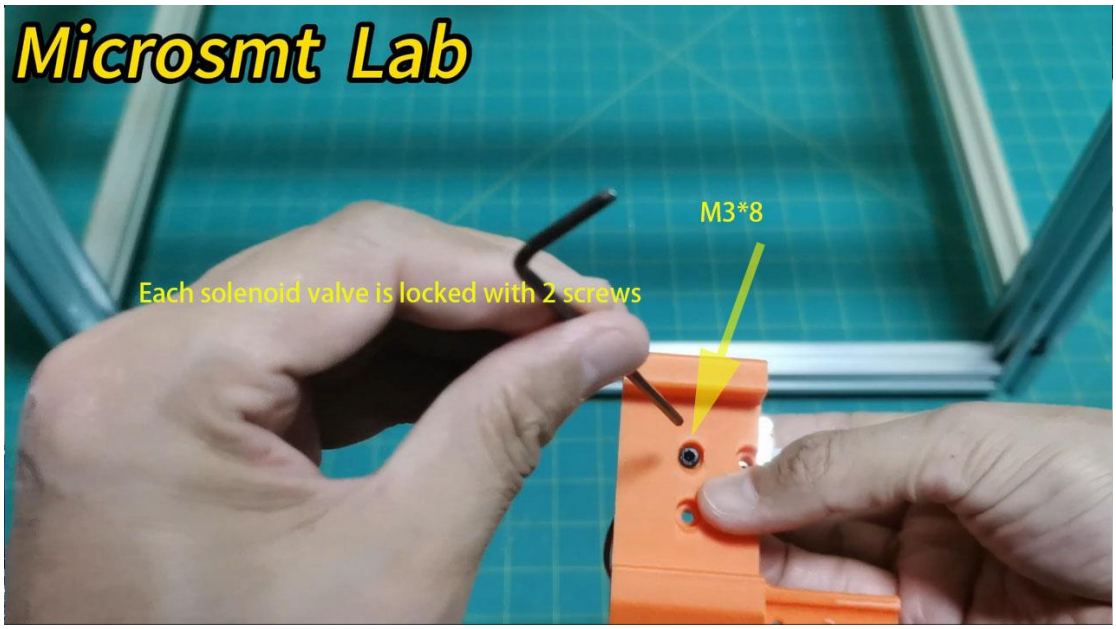
Please also refer to the assembly demonstration video. Can be completed faster.

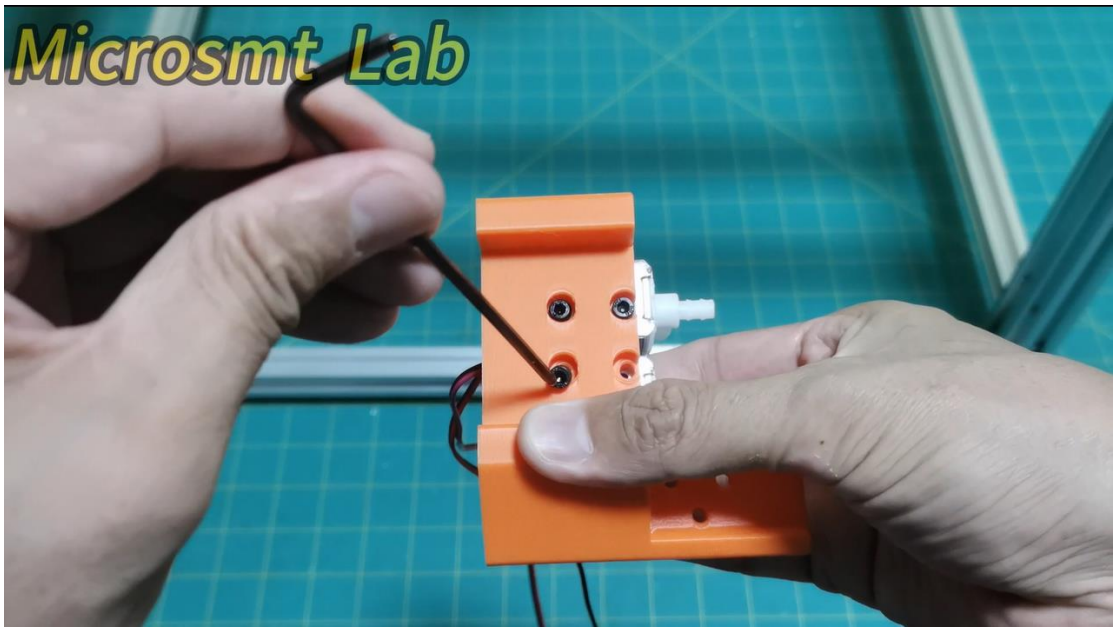
MicrosmtLab

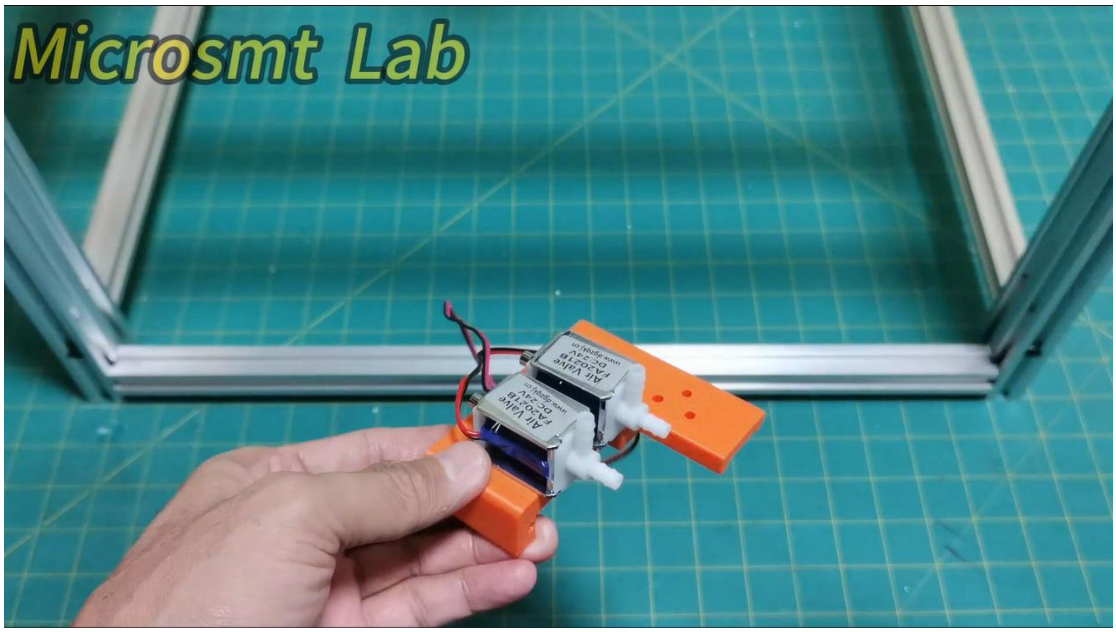
www.microsmt.com.cn

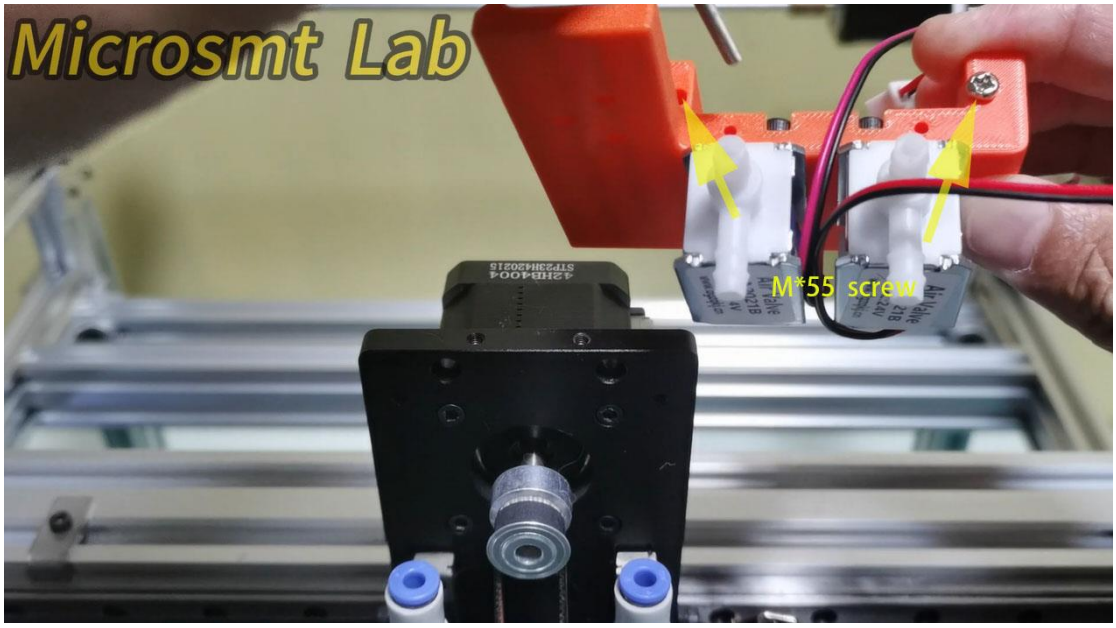
Step 1：Print Z-axis motor cover plate and Install solenoid valve.

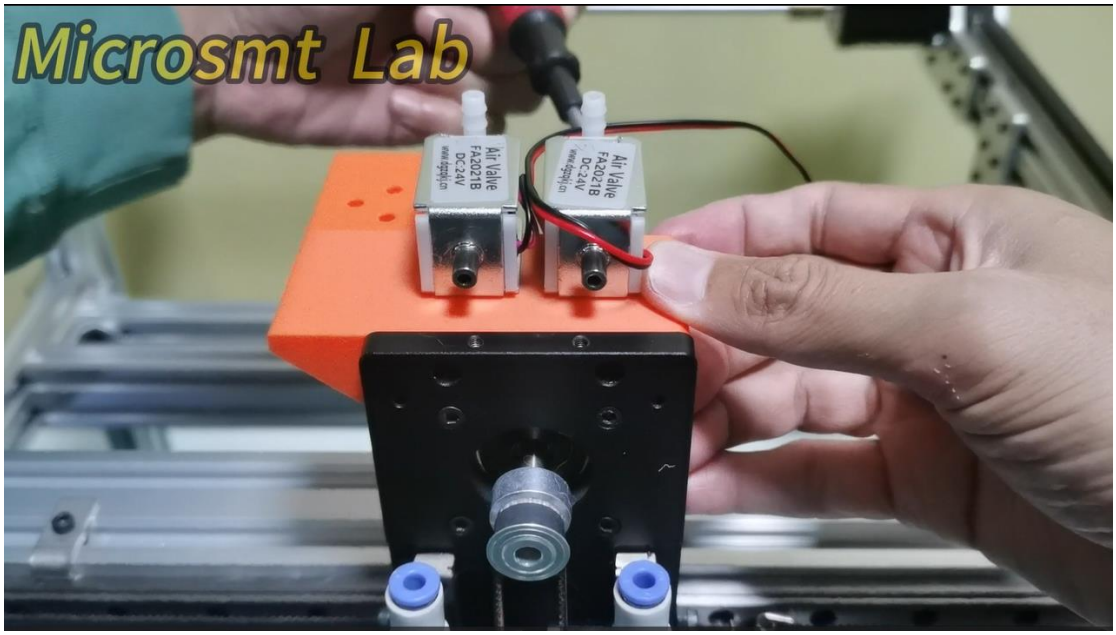




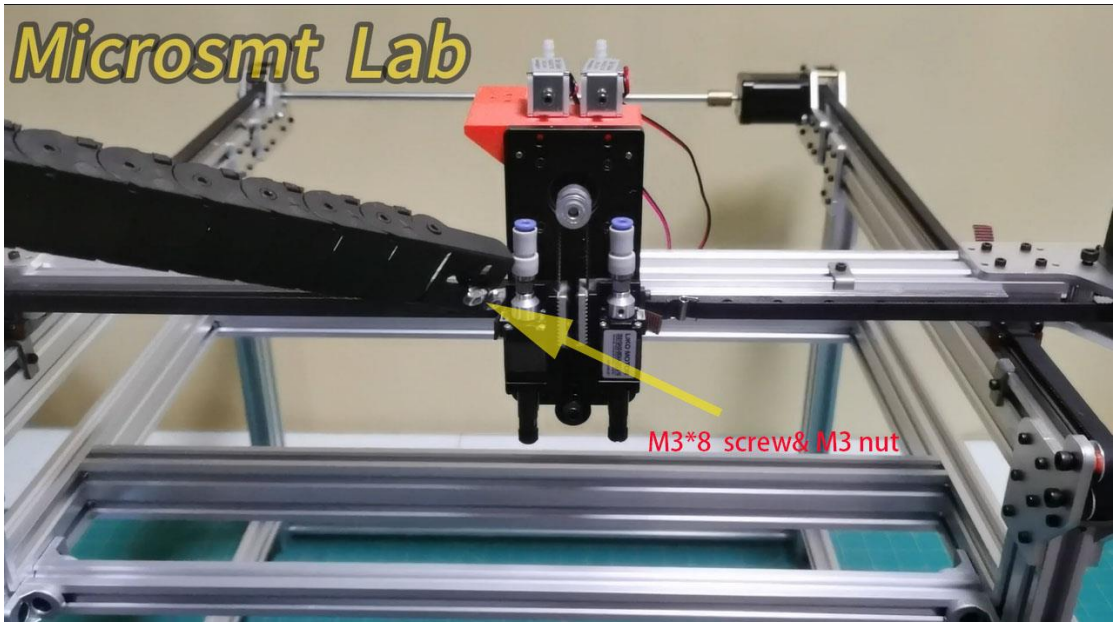

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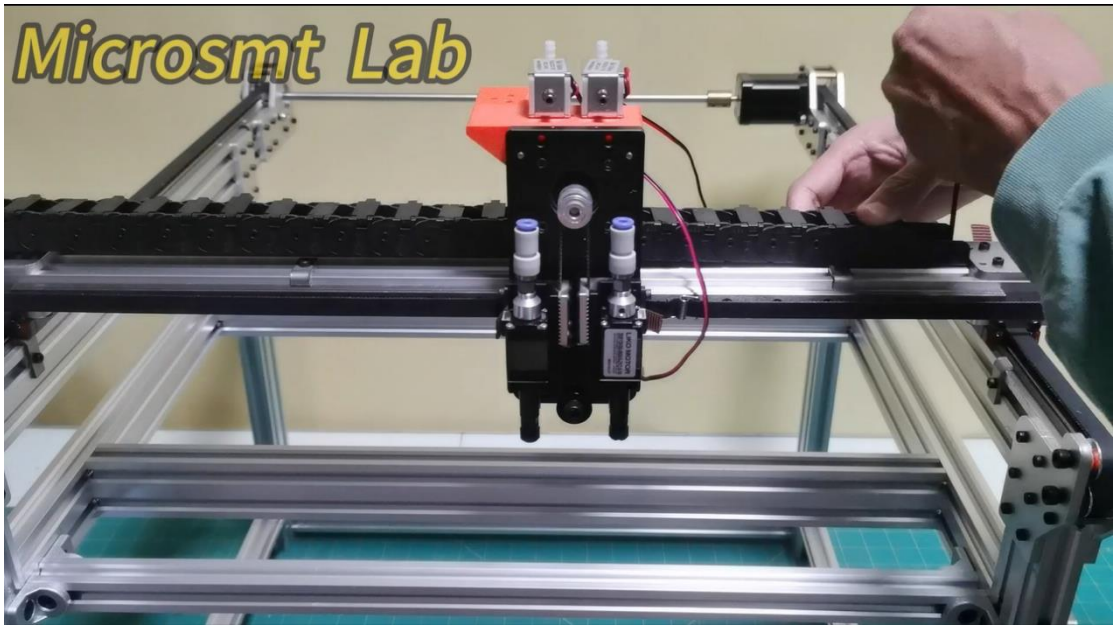

Microsmt Lab

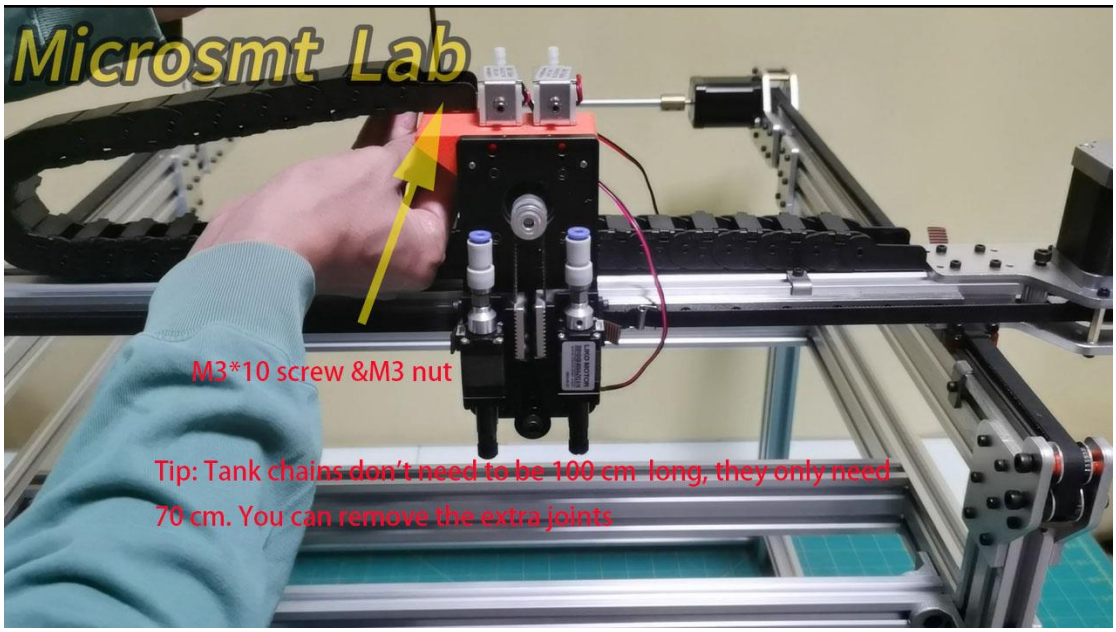

Microsmt Lab
M*55 screw


Microsmt Lab
Air Valve
FA2021B
DC:24V
www.sppkjx.cn

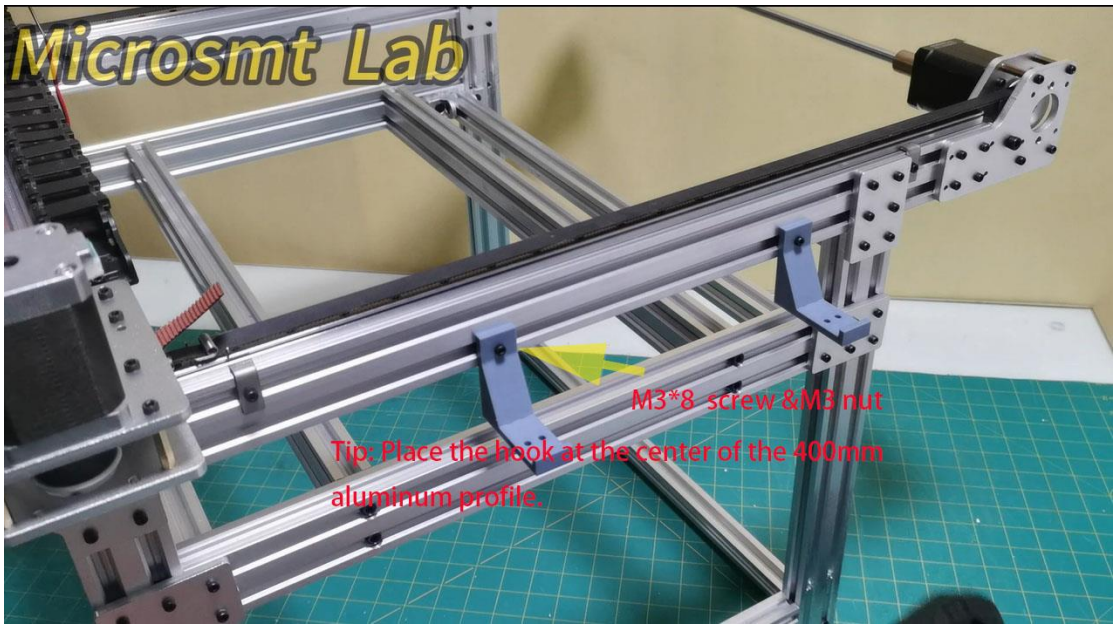
Step 2：Install x-axis tank chain.

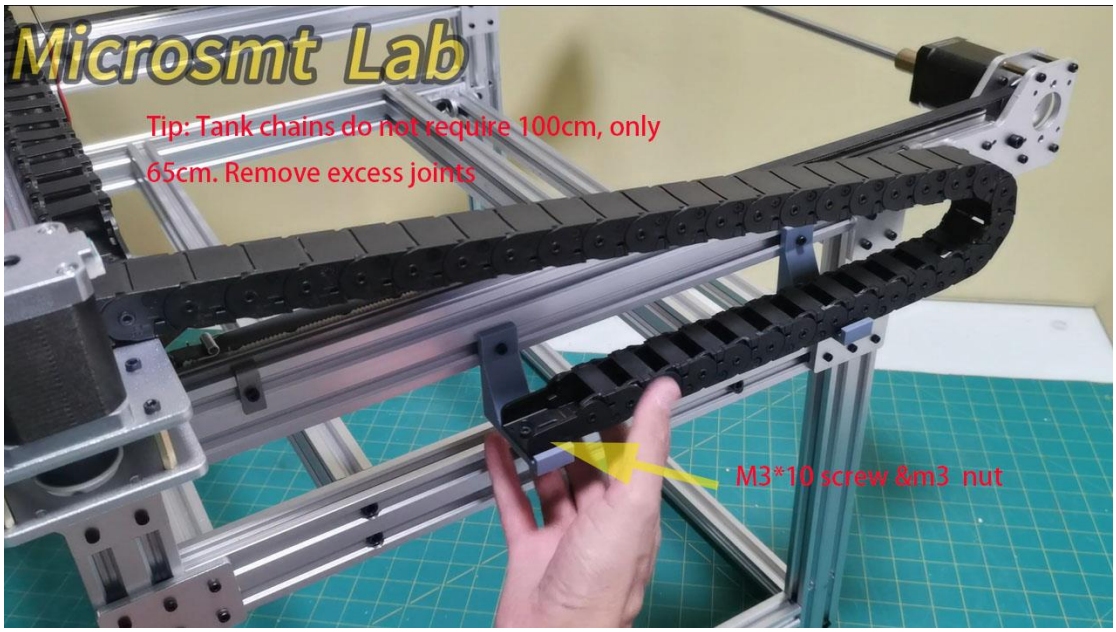


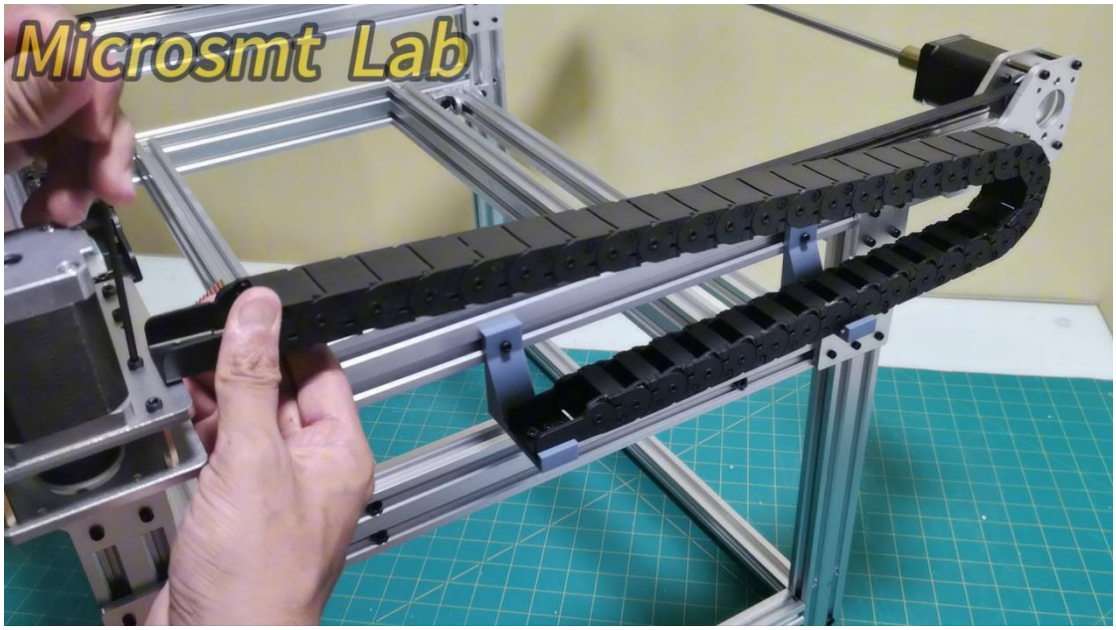




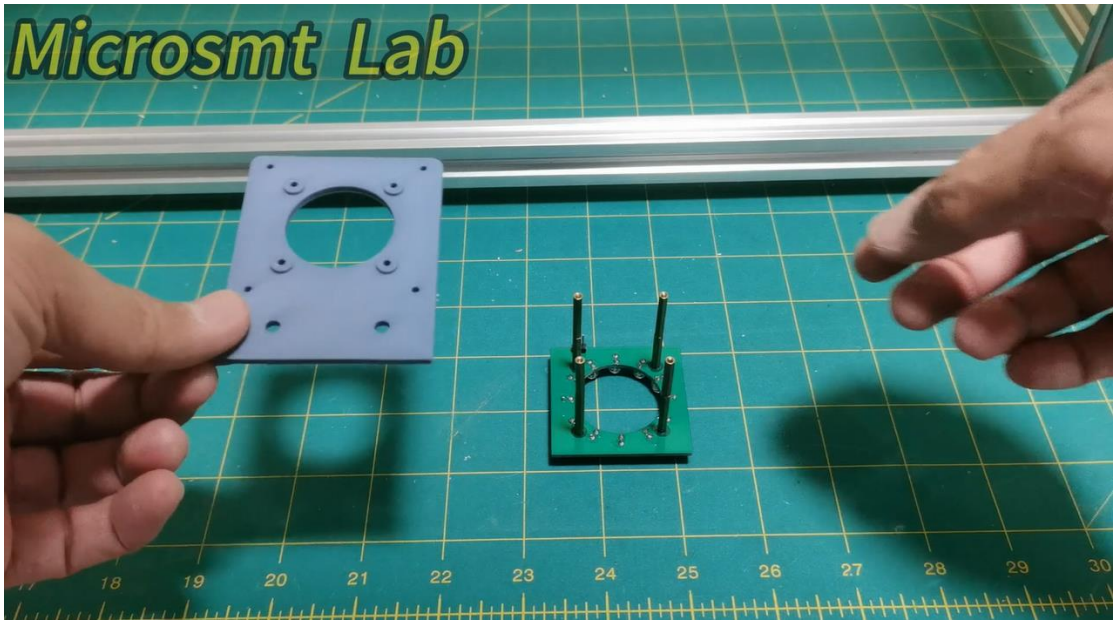
Step 3：Install y-axis tank chain.

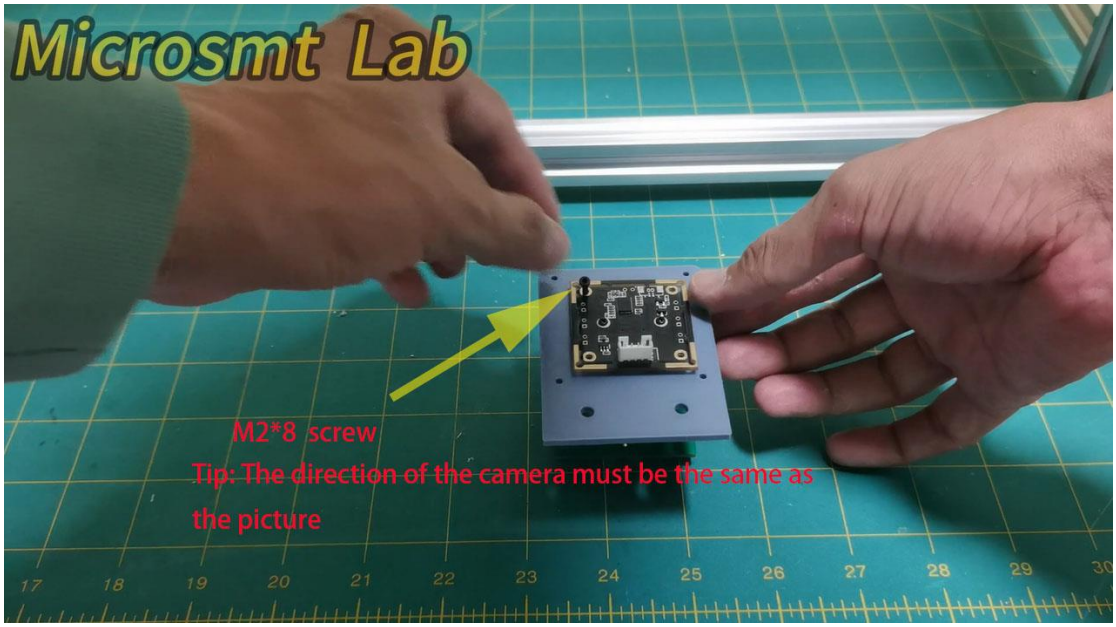


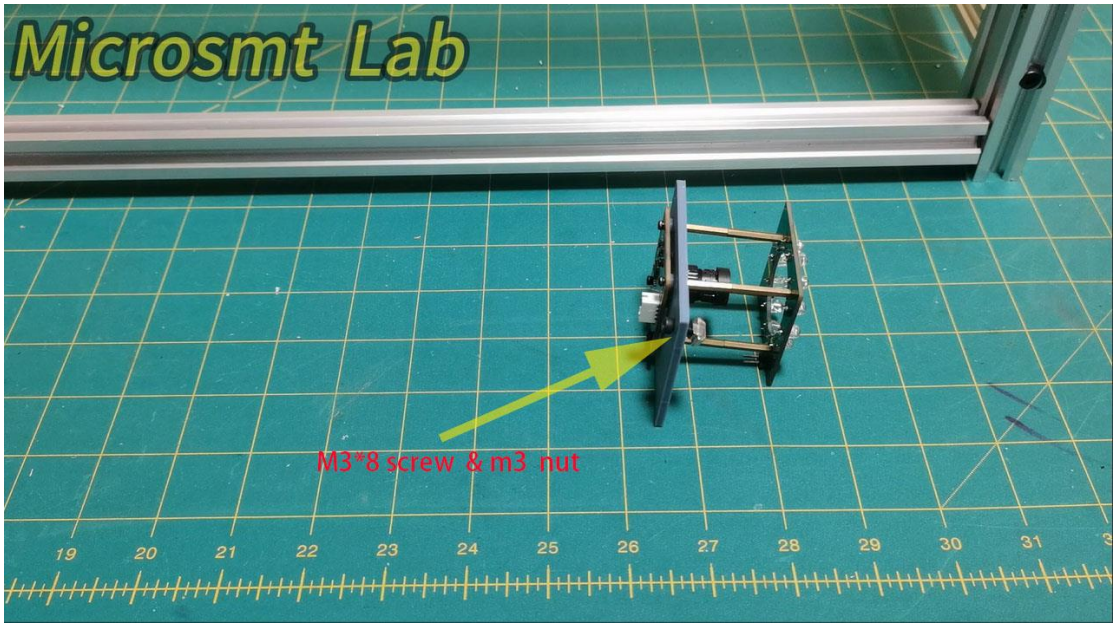

Microsmt Lab
Tip: Tank chains do not require 100cm, only
65cm. Remove excess joints
M3*10 screw &m3 nut

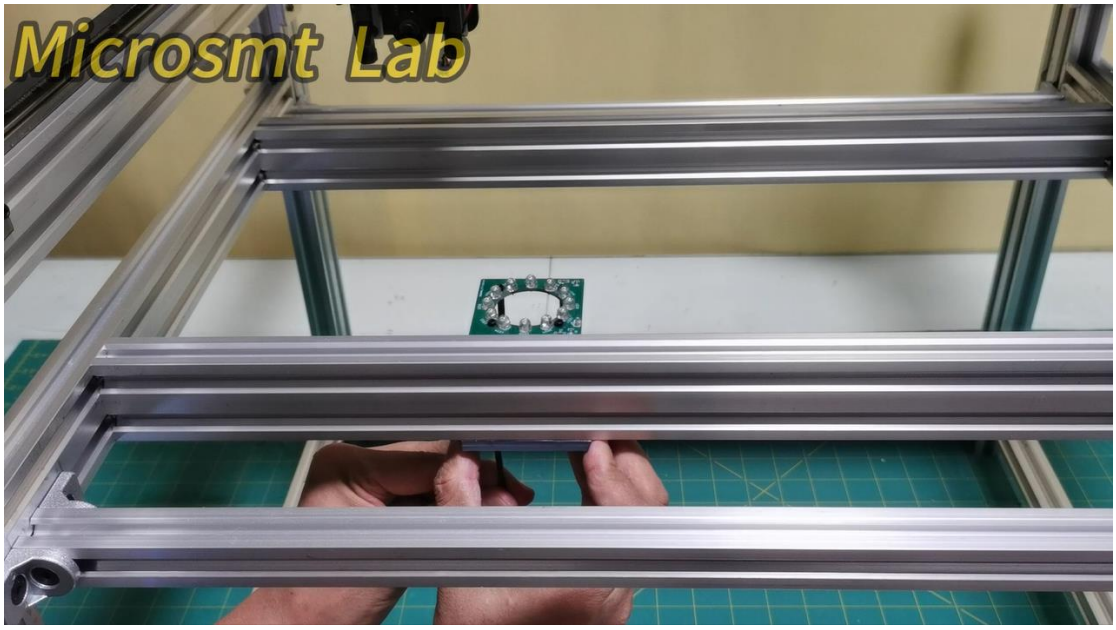

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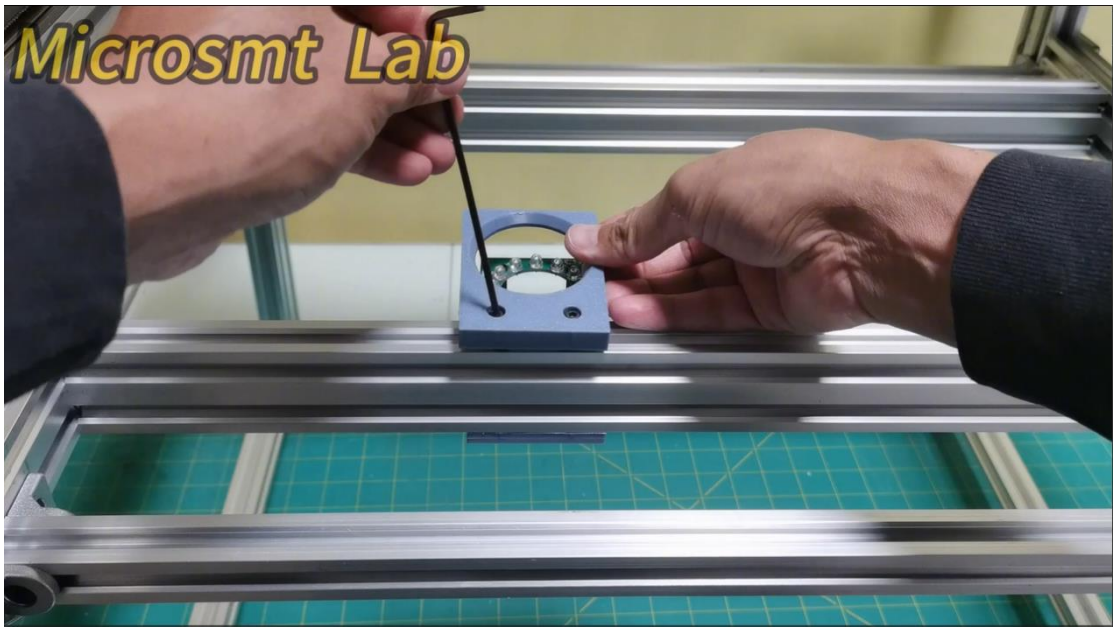
Step 4：Install the bottom camera.



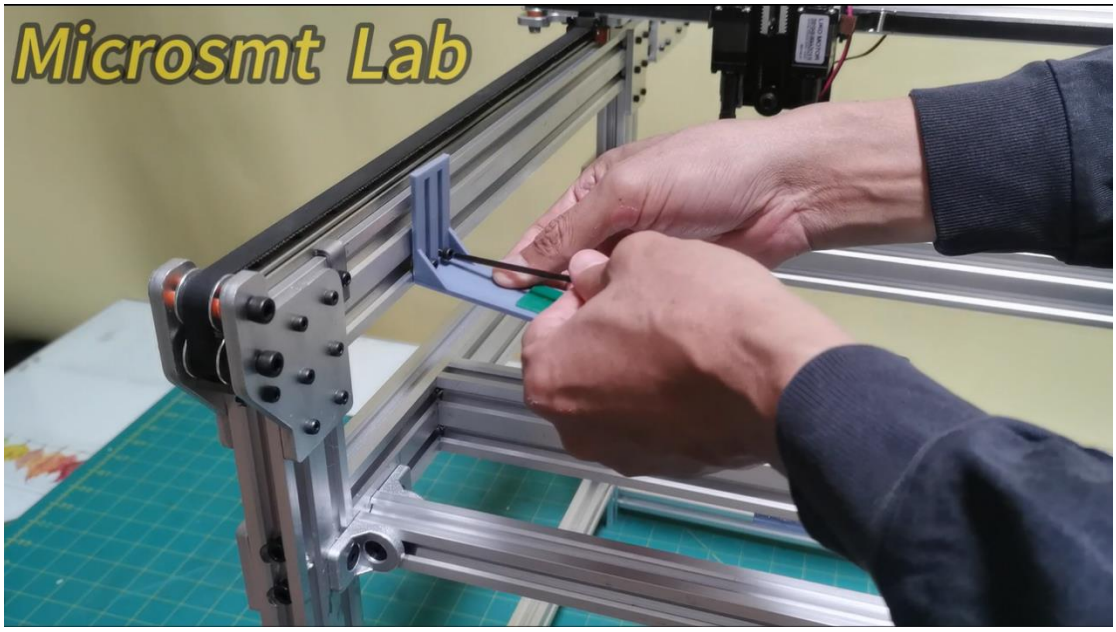



Microsmt Lab
M3*8 screw & m3 nut
19
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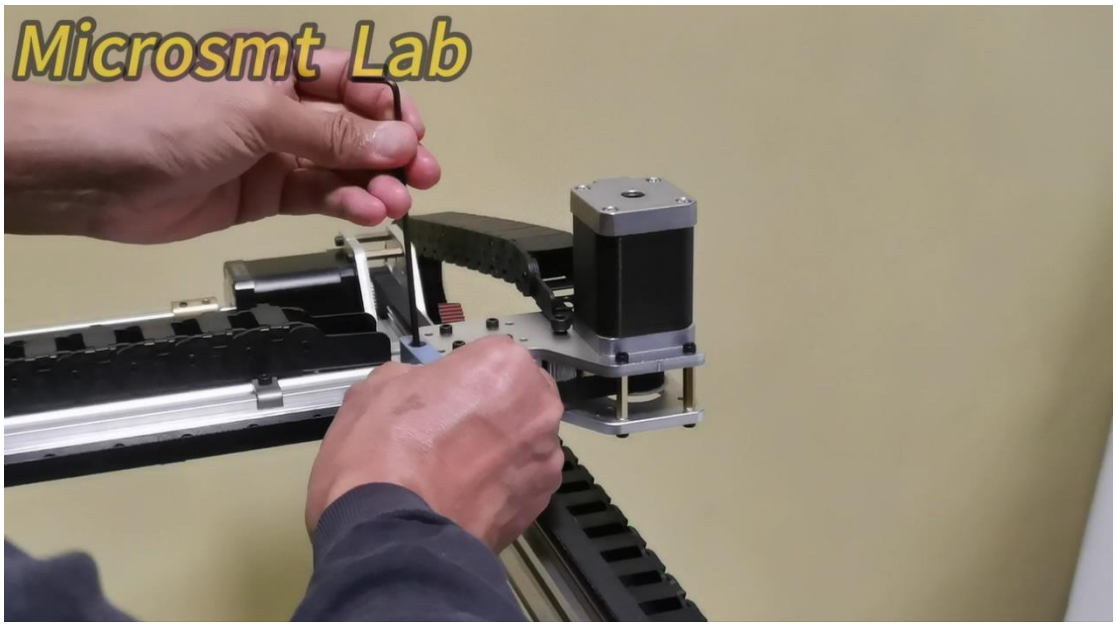

Microsmt Lab

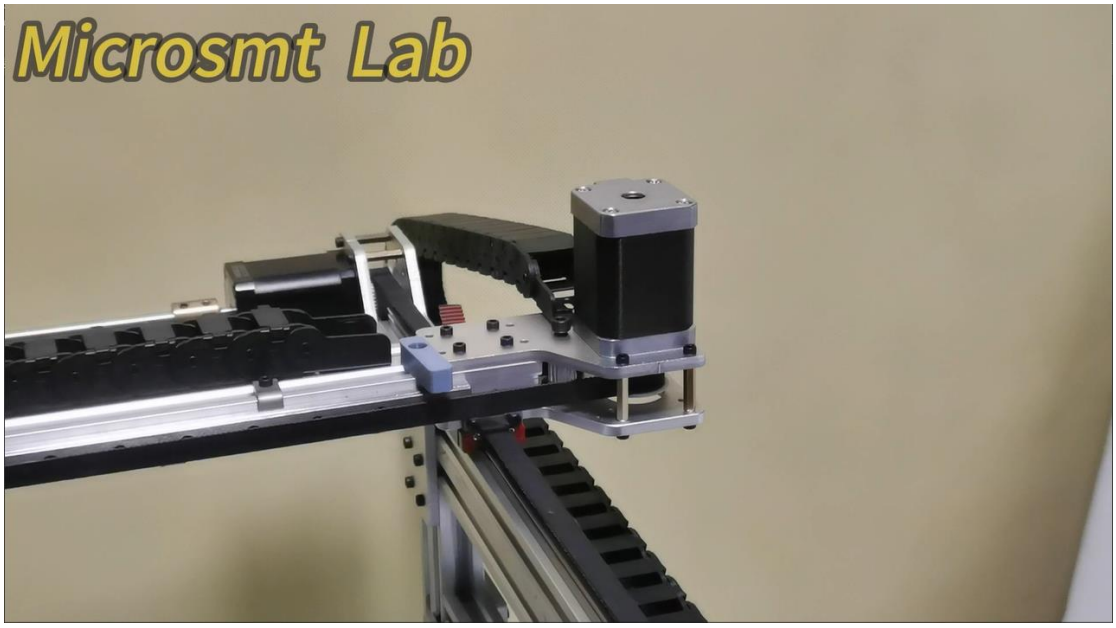


Step 5：Install Origin hook by M3*8 screw and m3 nut. There are two types of 3D printed parts, you can choose freely. The height of the homing point must be the same as the height of the PCB.

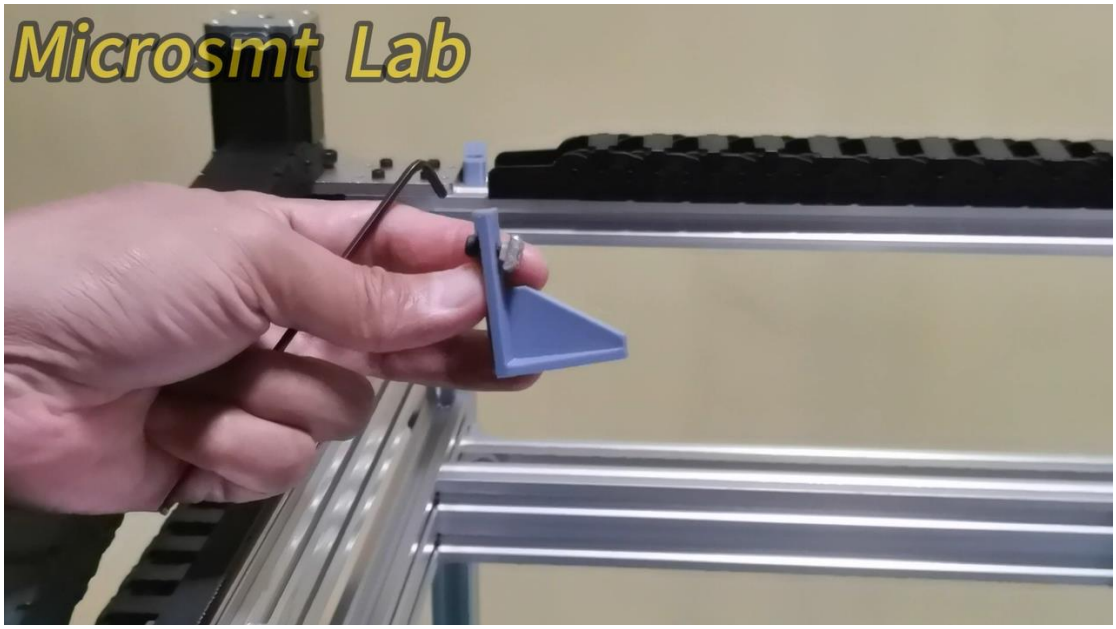


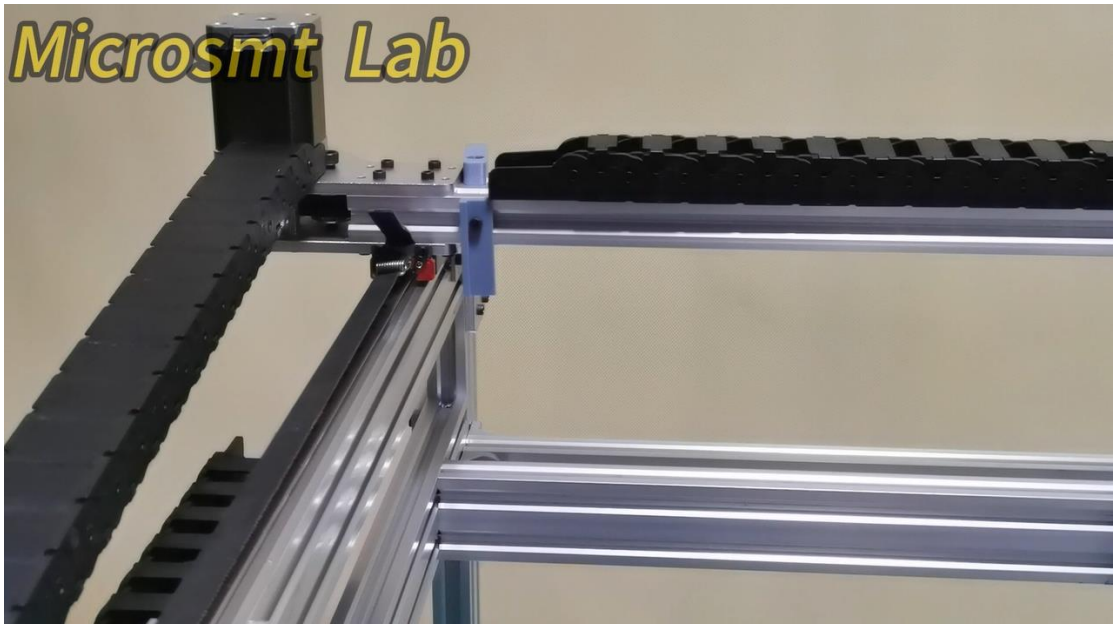
Step 6：Install　X-axis anti-collision by m3*8 screw and m3 nut.



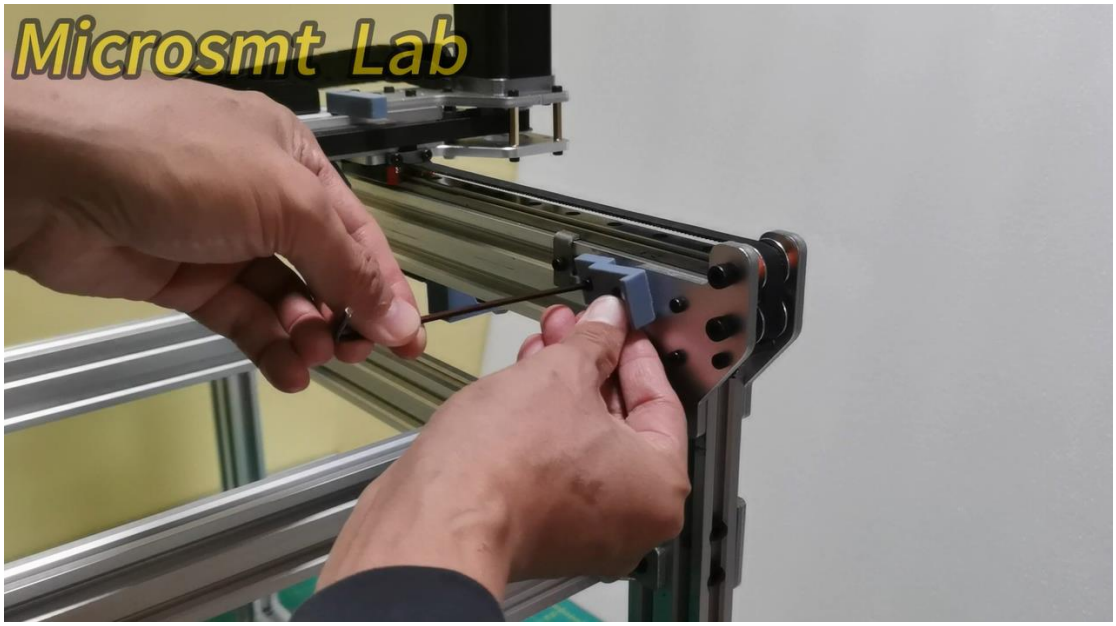


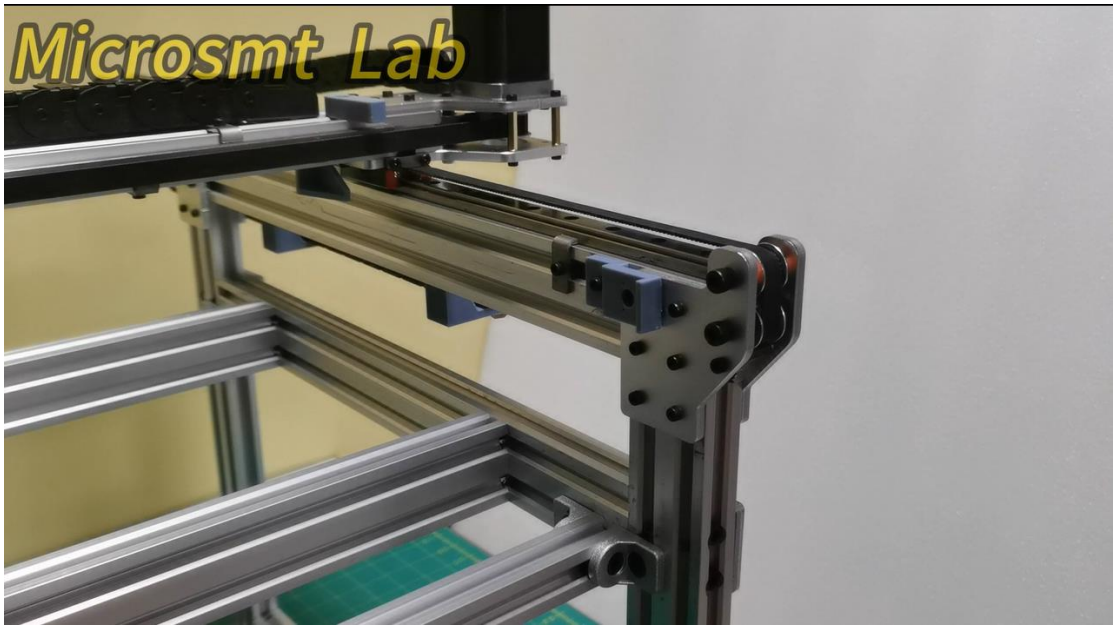
Step 7：Install Y-axis endstop striker by m3*8 screw and m3 nut.



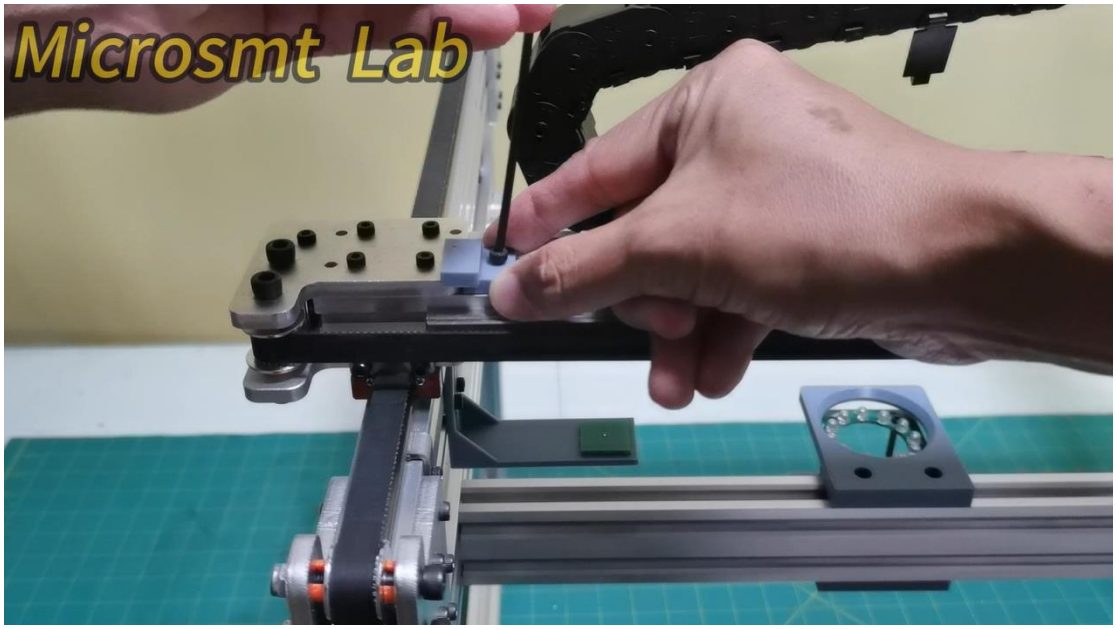


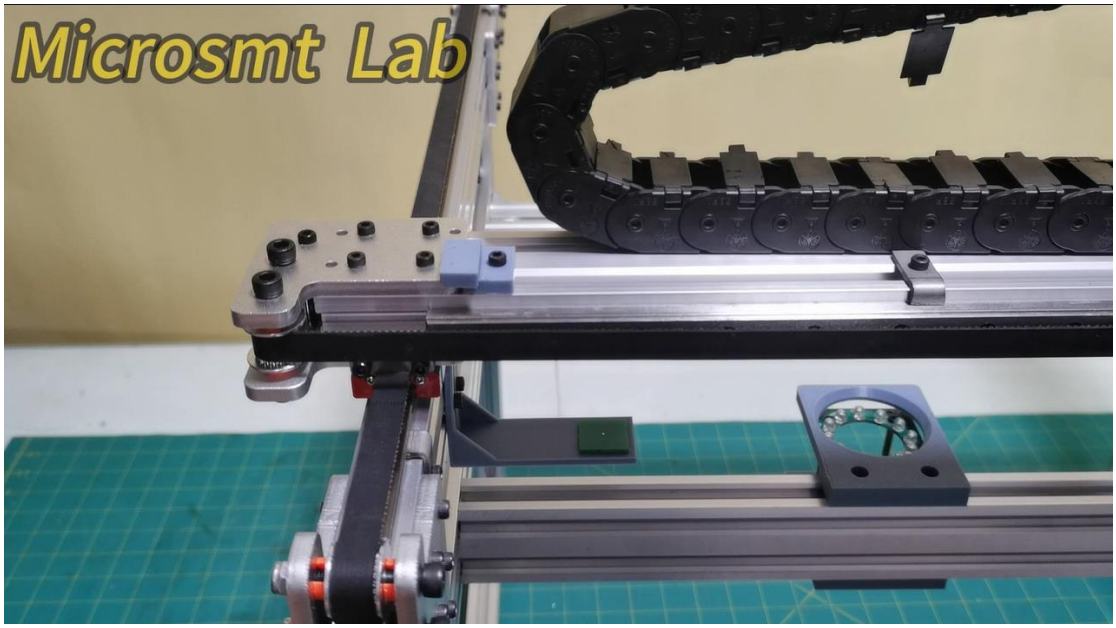
Step 8：Install Y-axis endstop switch Fixing plate by m3*8 screw and m3 nut.





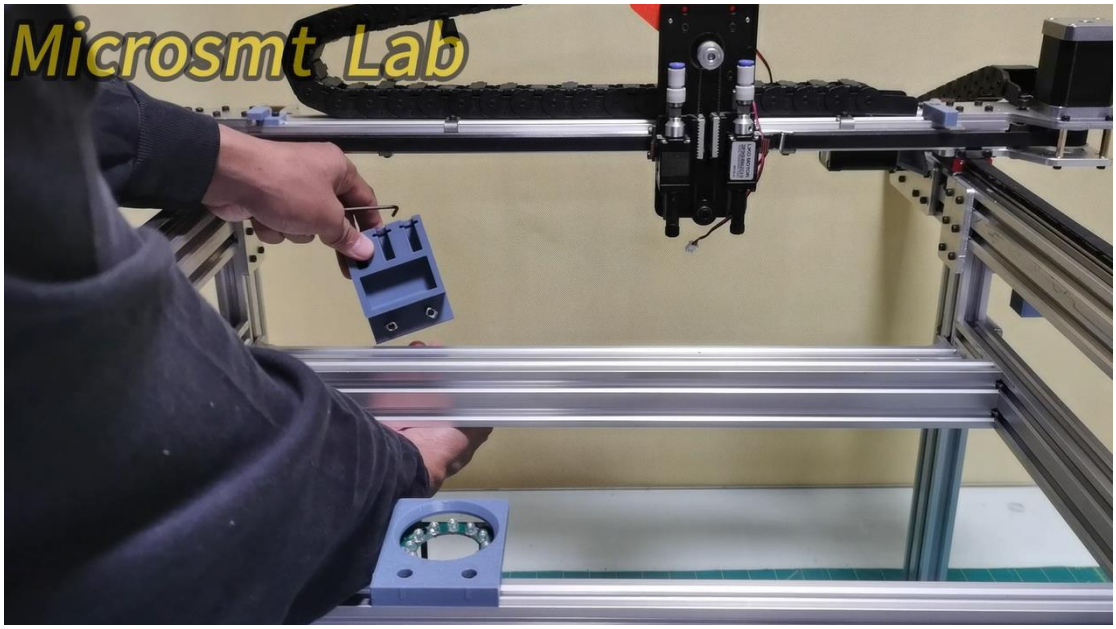
Step 9：Install X-axis endstop switch Fixing plate by m3*8 screw and m3 nut.

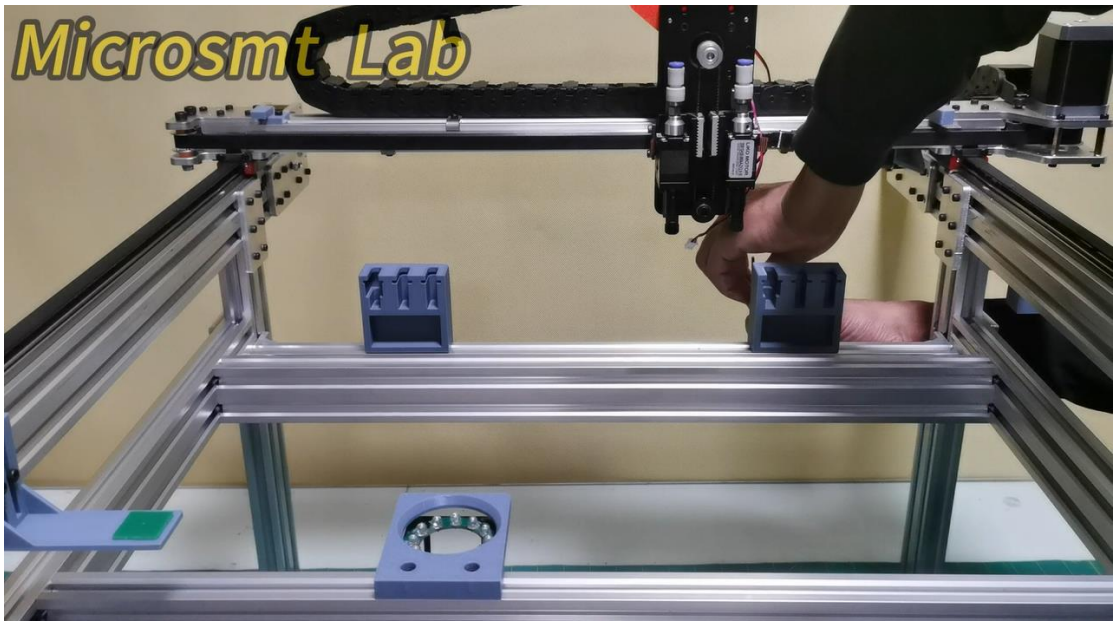




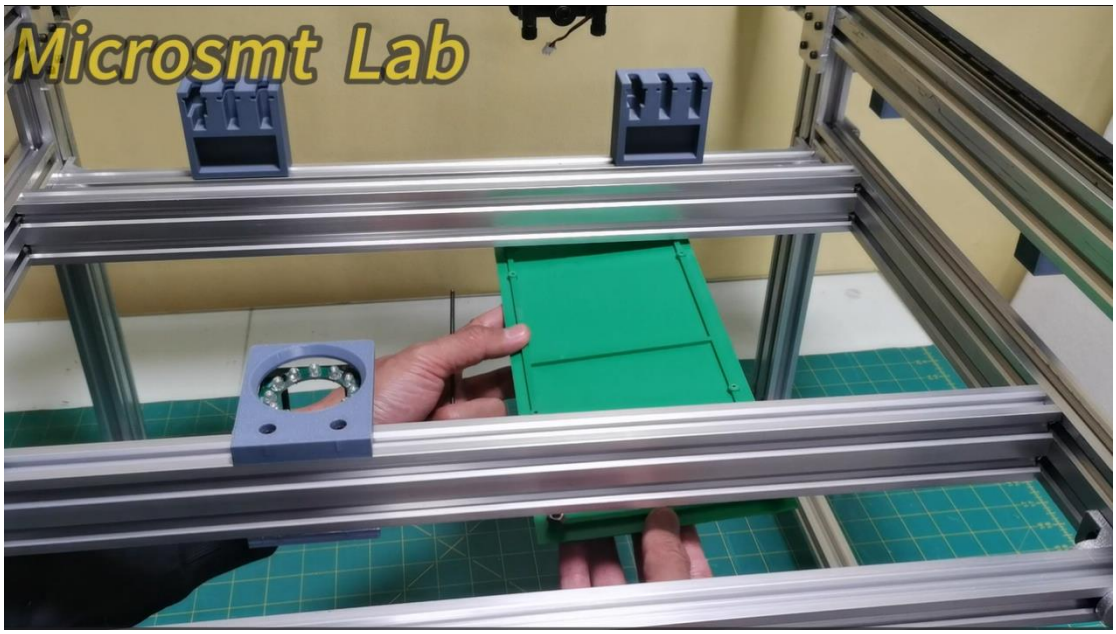
Step 10： Install CP45 nozzle rack by m3*8 screw and m3 nut.

You can delete the excess parts and fix them on both sides of the black platform with double-sided tape, which can provide more space for the bing feeder.





Step 11：Install motherboard fixing plate m3*8 screw and m3 nut.



Step 12：Install AC power socket,MW Switching Power Supply, vacuum pump.



Attention: Set the dip code of the switch power supply to 110V or 220V according to your country's voltage.



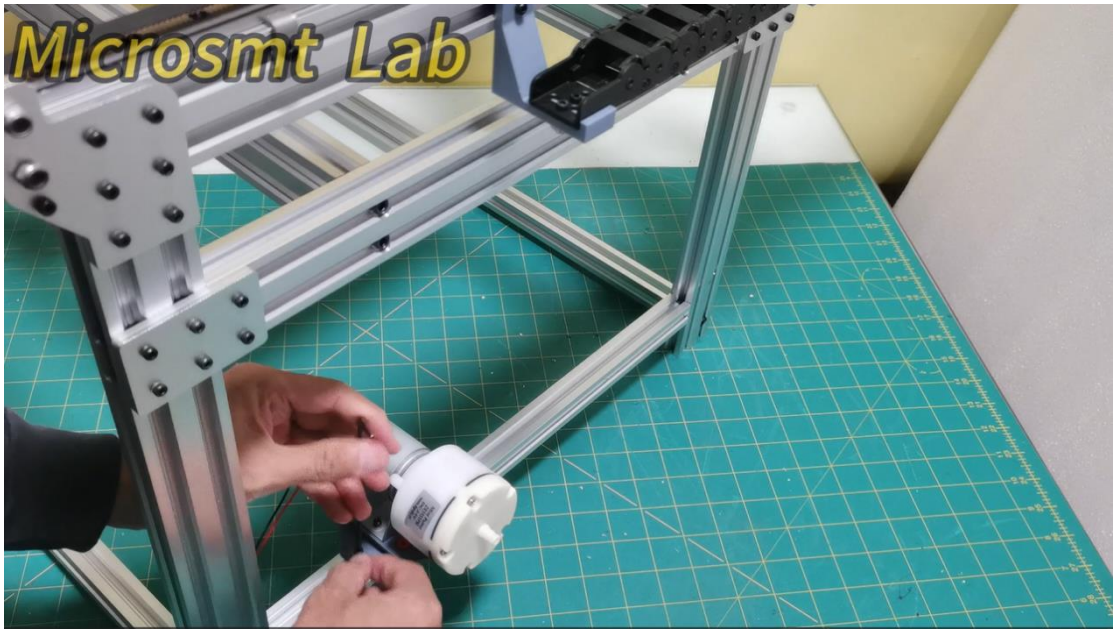


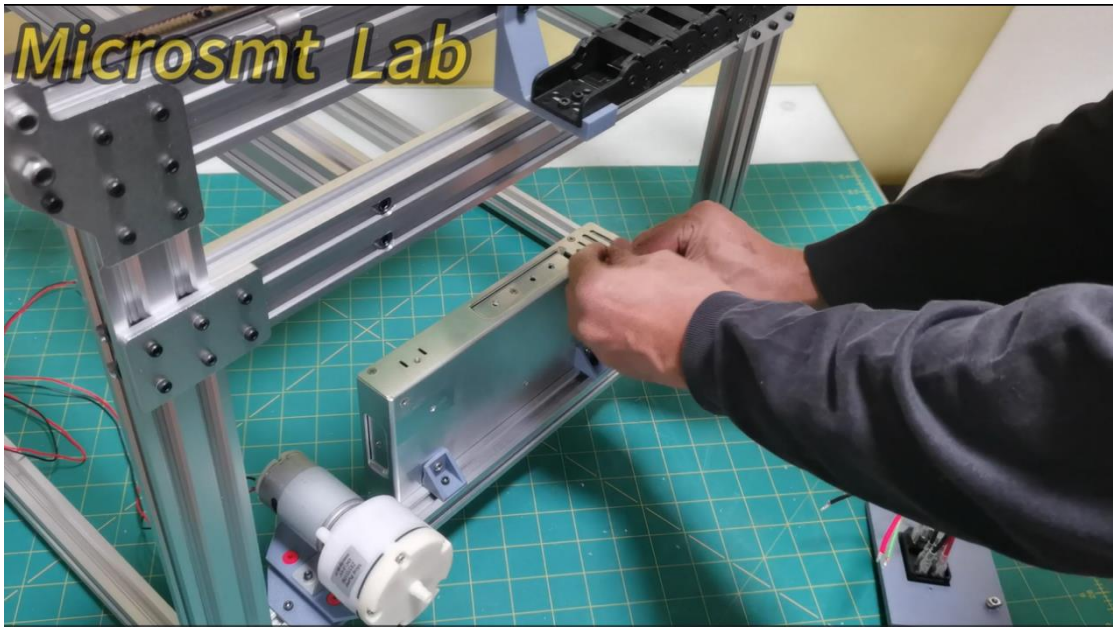

Microsmt Lab
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M5*10 screw

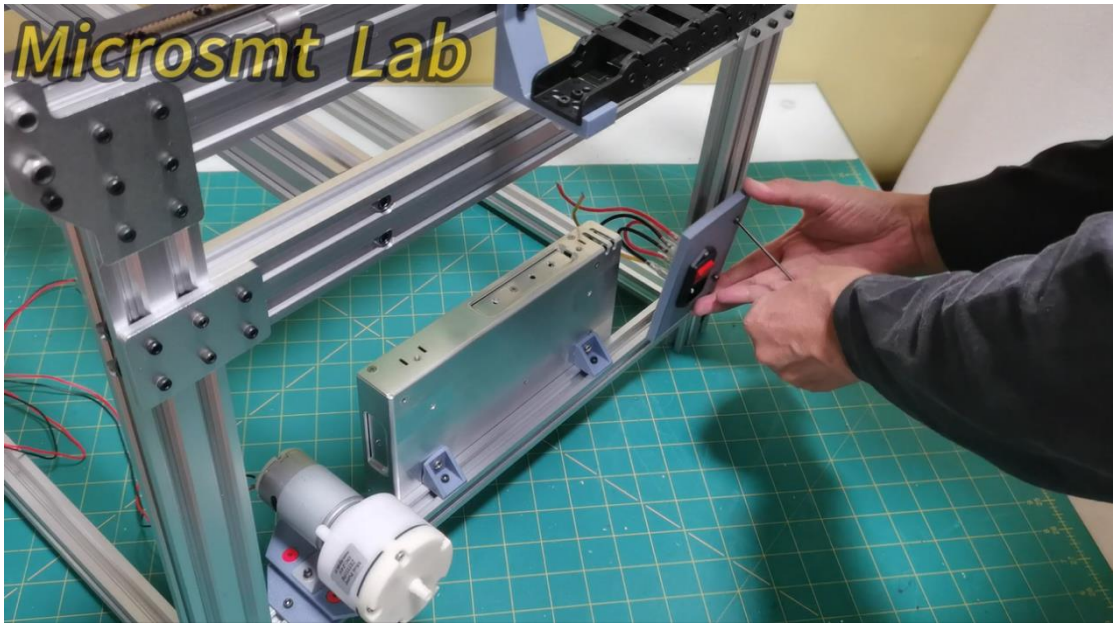

Microsmt Lab
www.microsmt.com.cn


Microsmt Lab
www.microsmt.com.cn

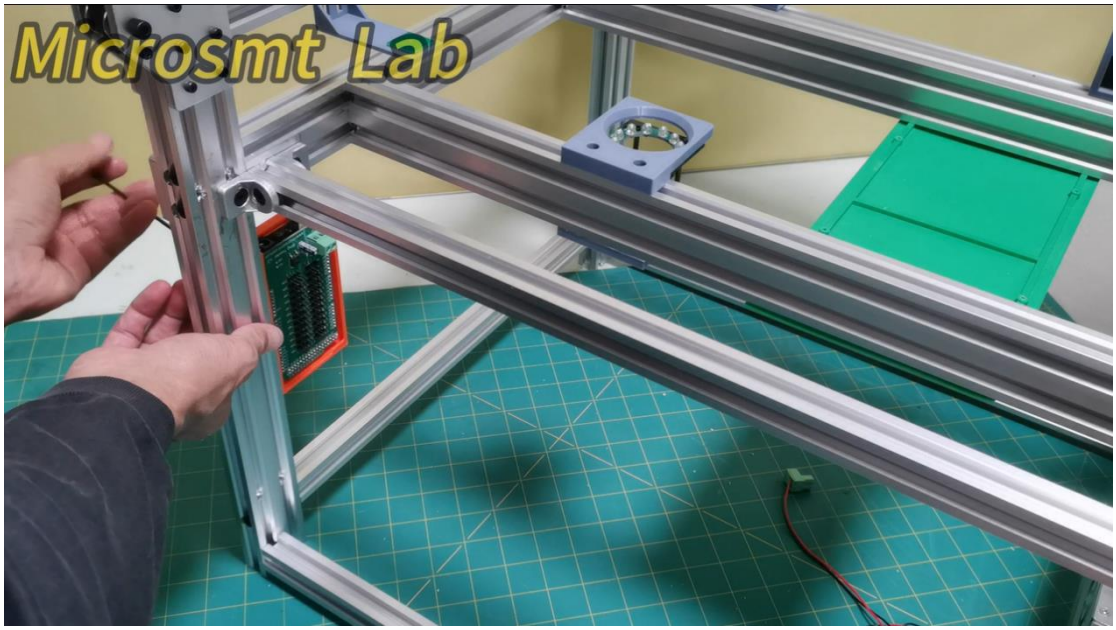

Microsmt Lab
www.microsmt.com.cn
M3*8 screw &m3 nut


Microsmt Lab

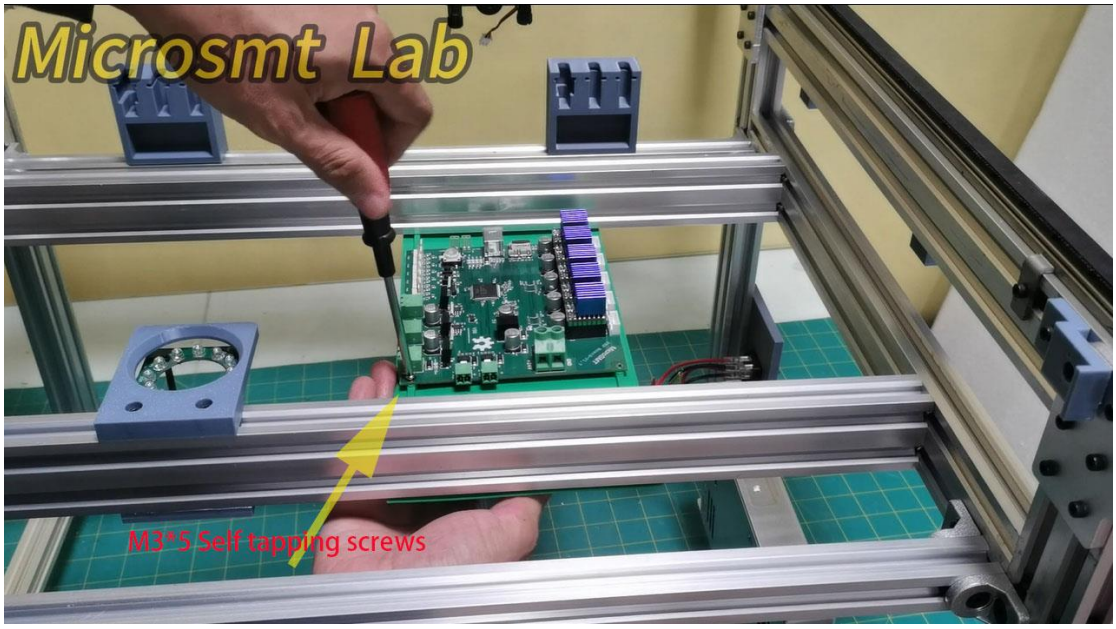

Microsmt Lab



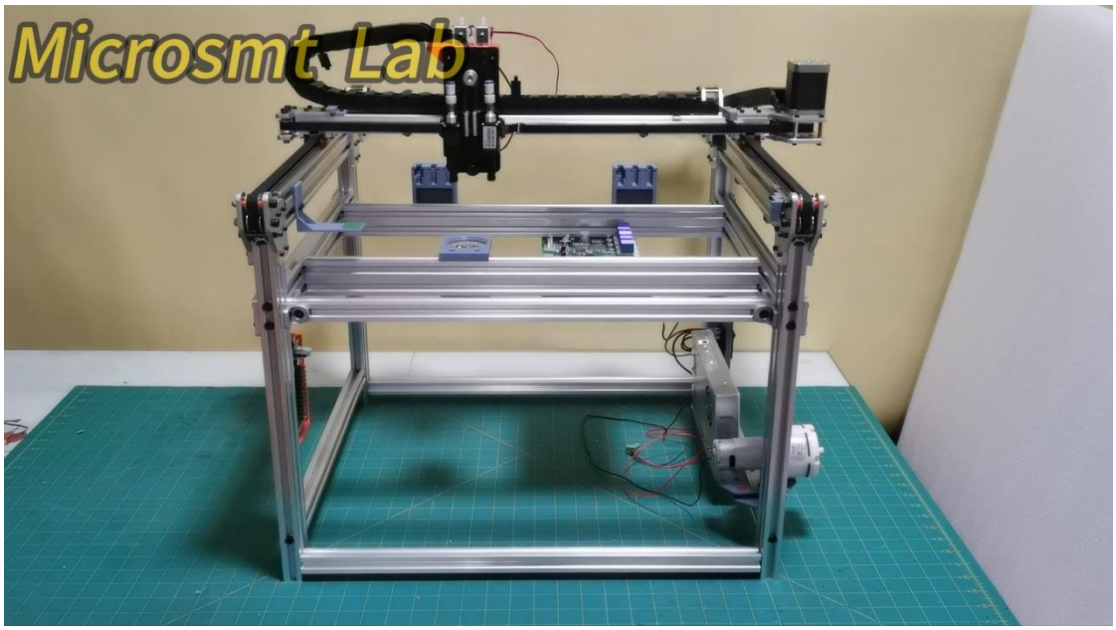
Step 13：Install feeder controller MEGA2560



Step 14：Install motherboard.



Preparing to install cable and air tube.



Step 15: Install cable and air tube for mounter head.

Include:

Z motor cable 215cm * 1

N1 motor cable 215cm *1

N2 motor cable 215cm *1

Solenoid valve cable 215cm *2

Top LED cable 215cm*1

Top camera cable 200cm * 1

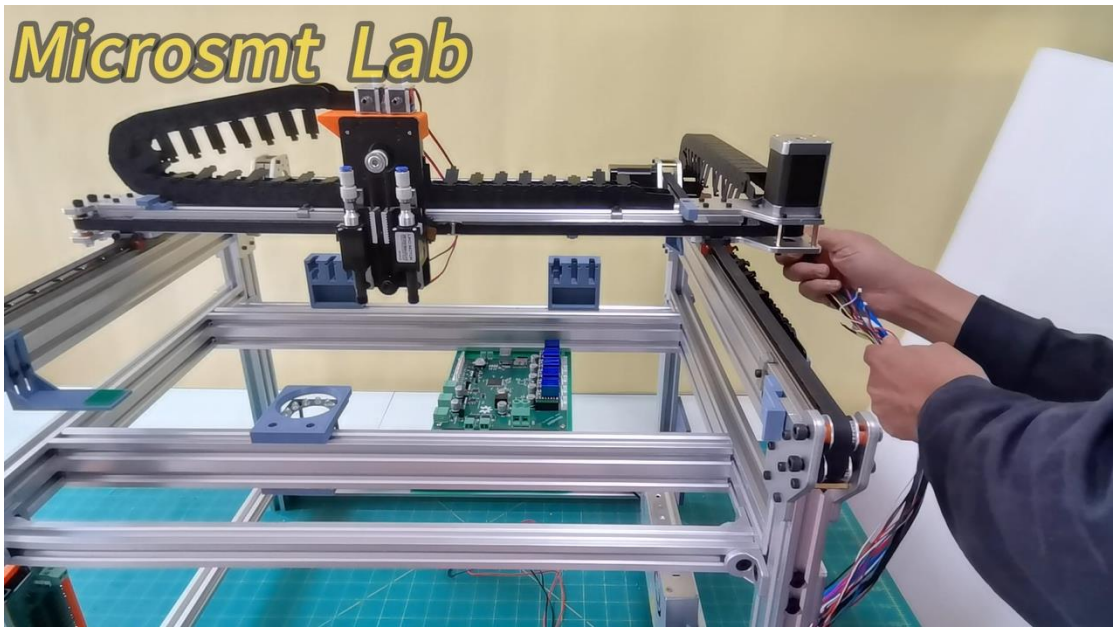
8mm air tube*1

Z-axis position sensor cable 215mm*1

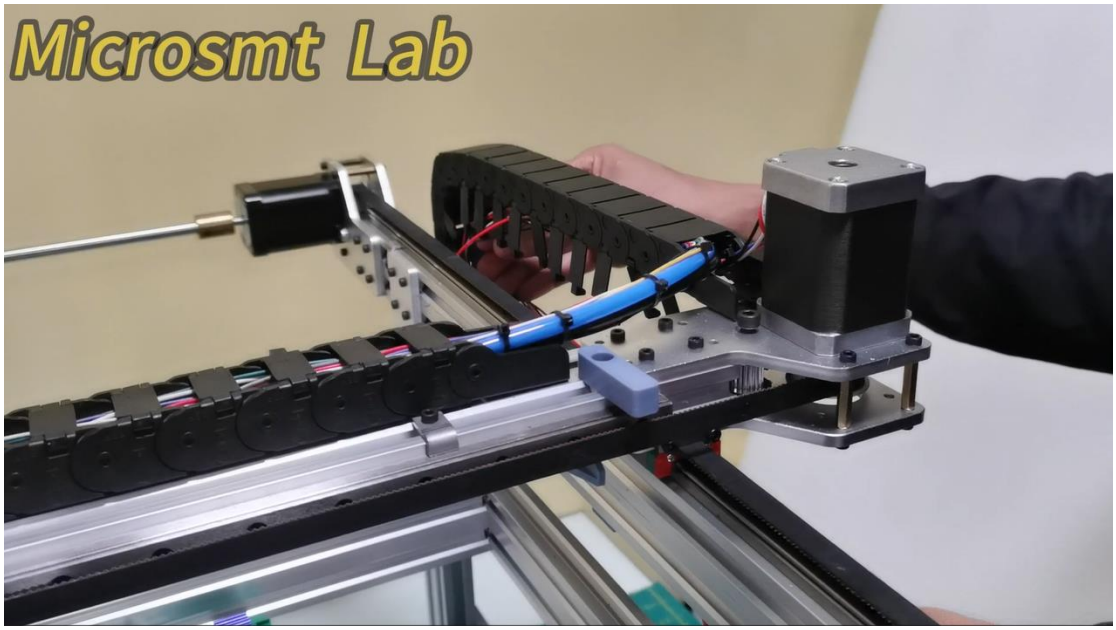
X-axis position sensor cable 215mm*1

Tip1: Please refer to the video to place cable.

Tip2: At the same time, put the 125cm long X-axis motor cable inside Y-axis tank chain.

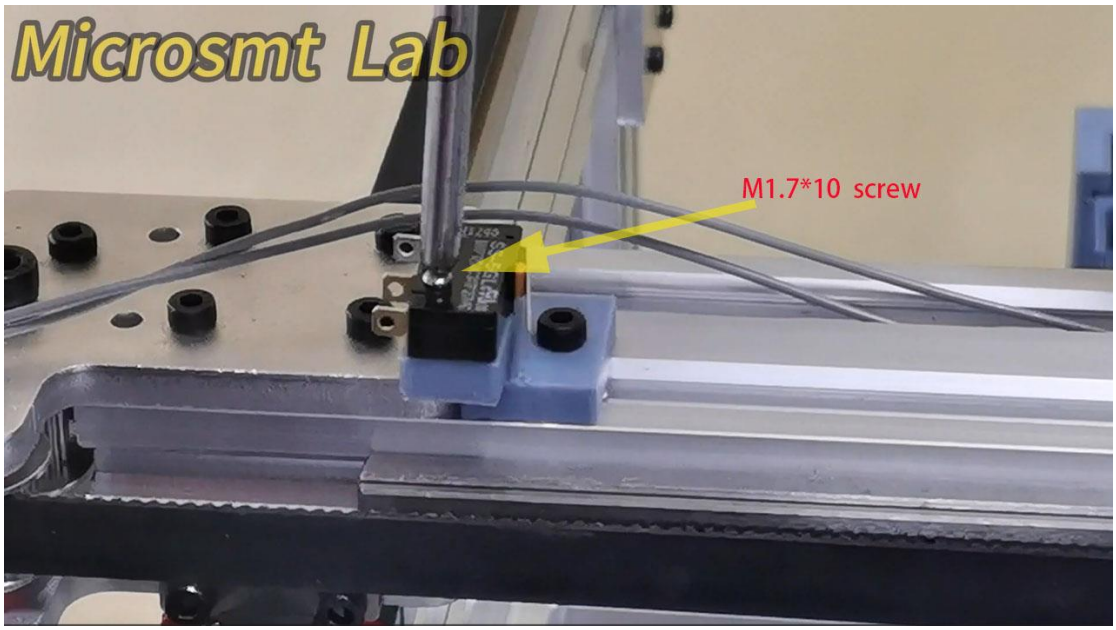


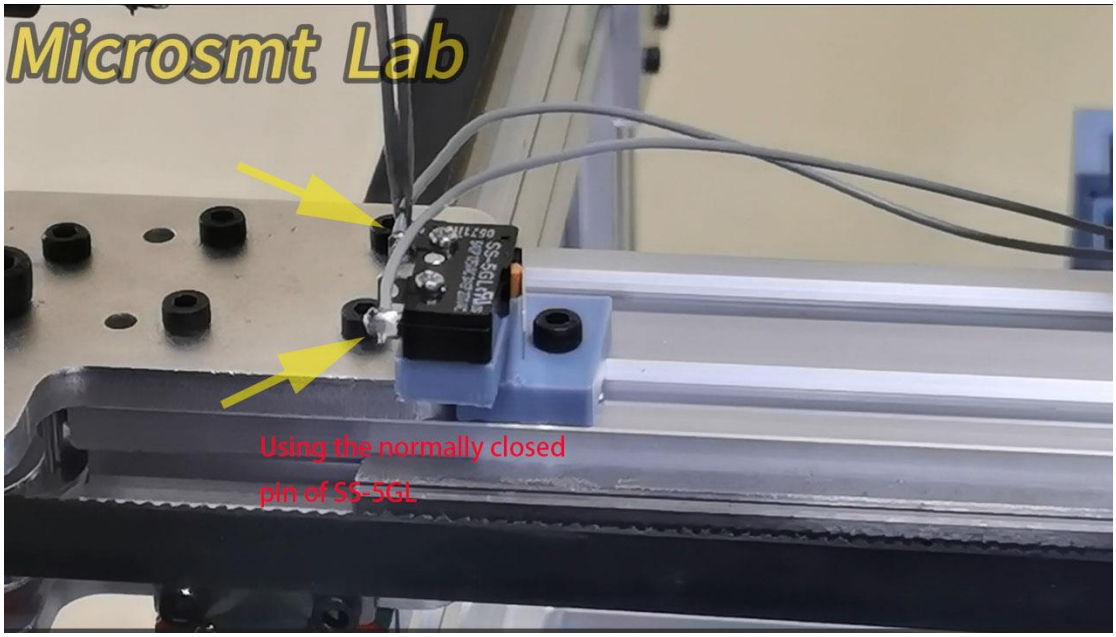
Step 16：Place the X-axis motor cable.



Step 17：Install the x-axis position sensor and its cable.

Tip: Please refer to the video to place cable.

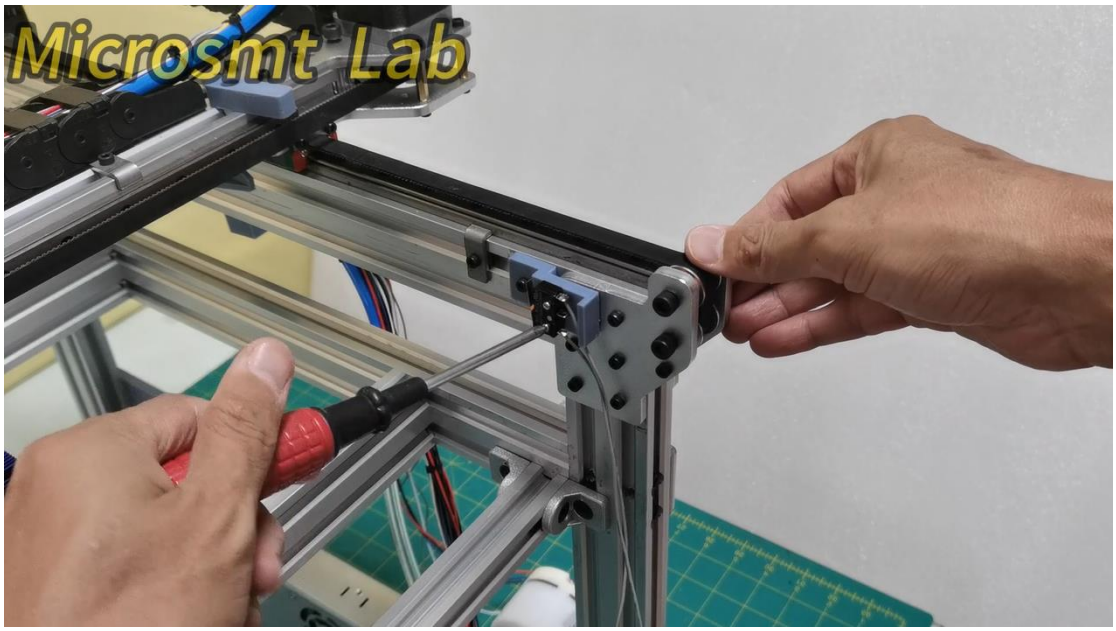




Step 18：Install the y-axis position sensor and its cable.

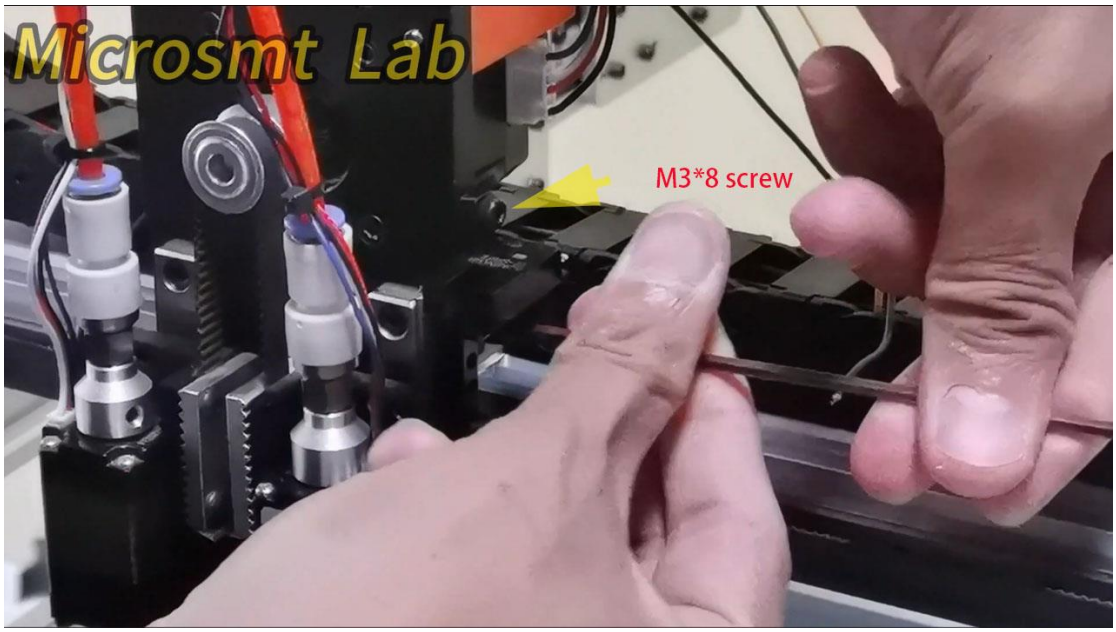
Tip1： Please refer to the video to place cable.

Tip2： Using the normally closed pin of SS-5GL.

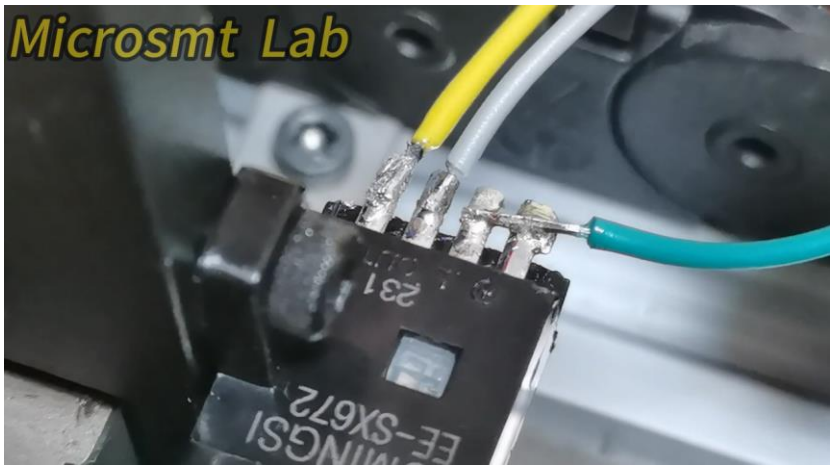


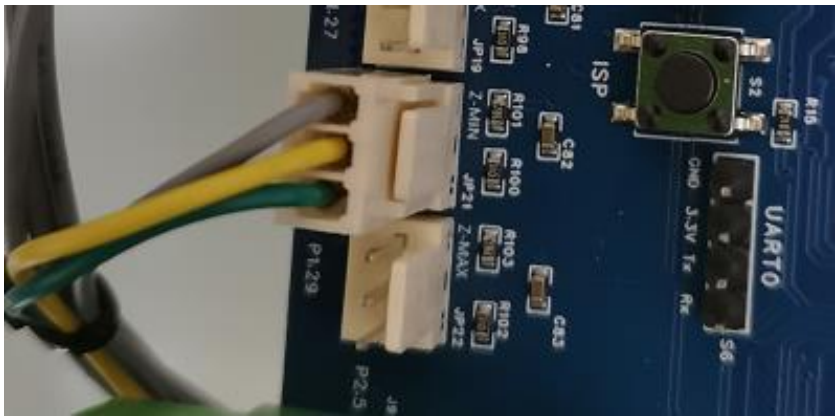
Step 19：Install the z-axis position sensor and its cable.

Tip1 : Please refer to the video to place cable.

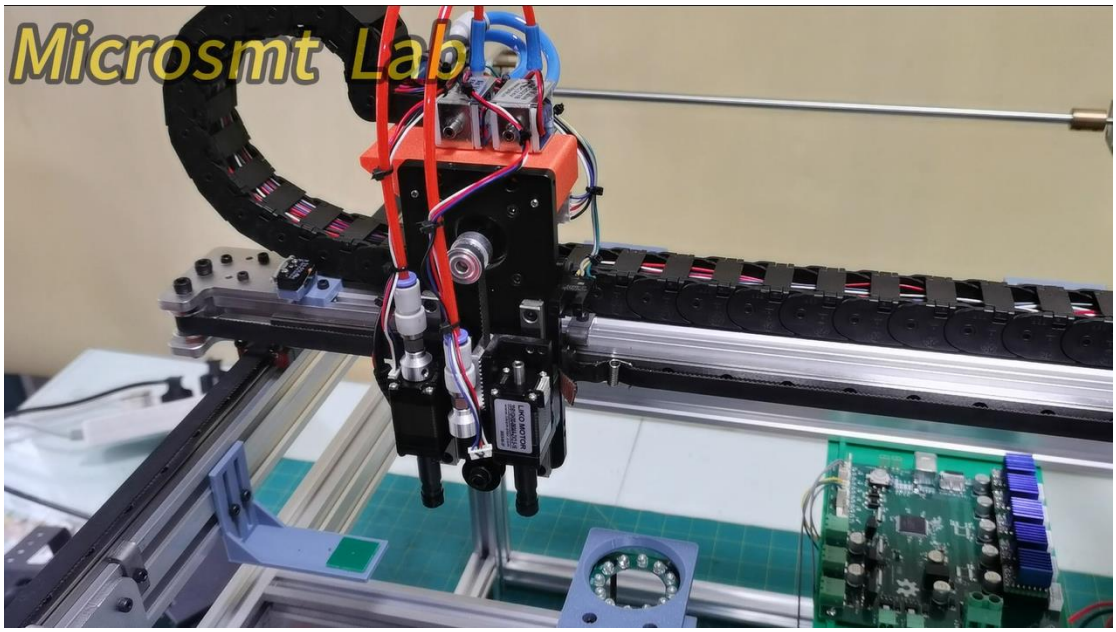


Tip2 : Please refer to the manual of motherboard to solder the EE-SX672 photoelectric switch, If you haven't read the motherboard manual, you won't be able to use it correctly.

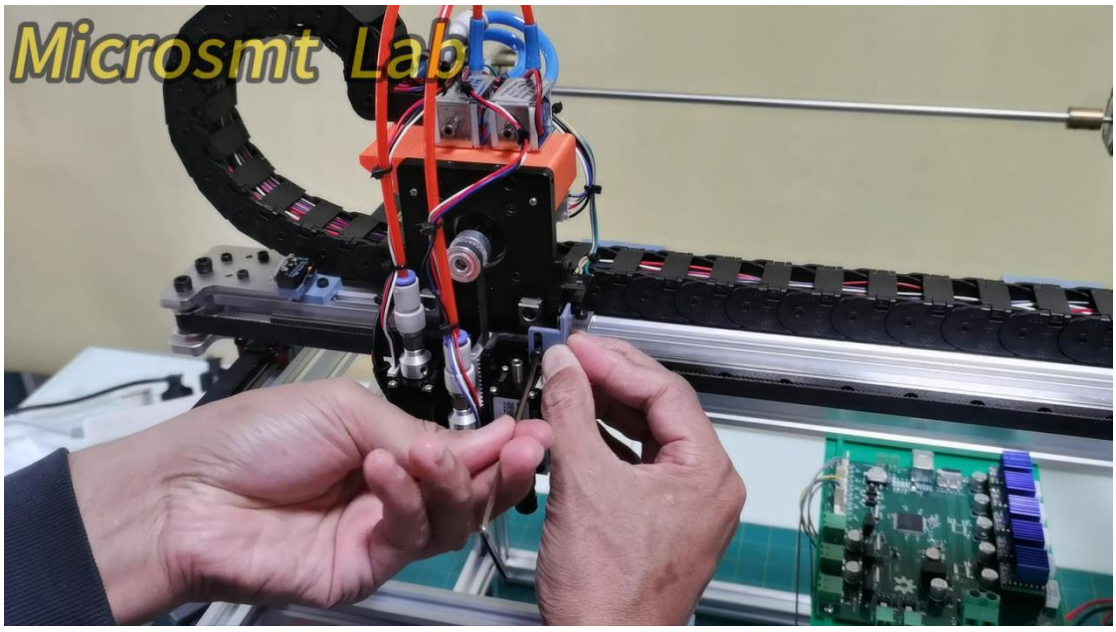




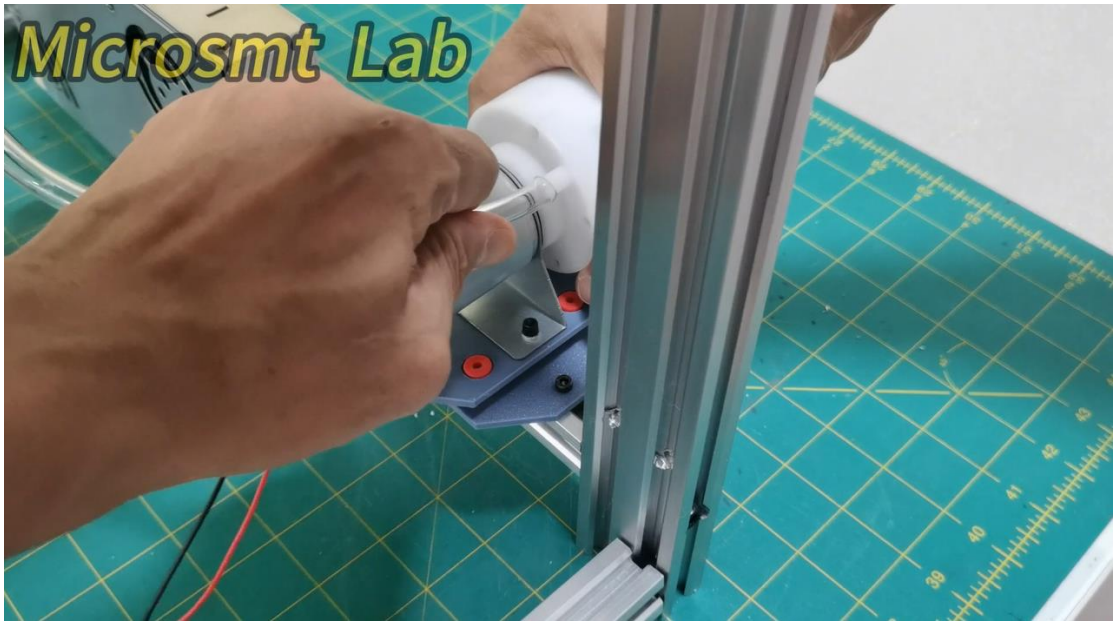
Step 20：Install Z-axis endstop striker.
Remove the rotating connection first



Fix it with M3 * 8 screws, It's height needs to be modified during debug.



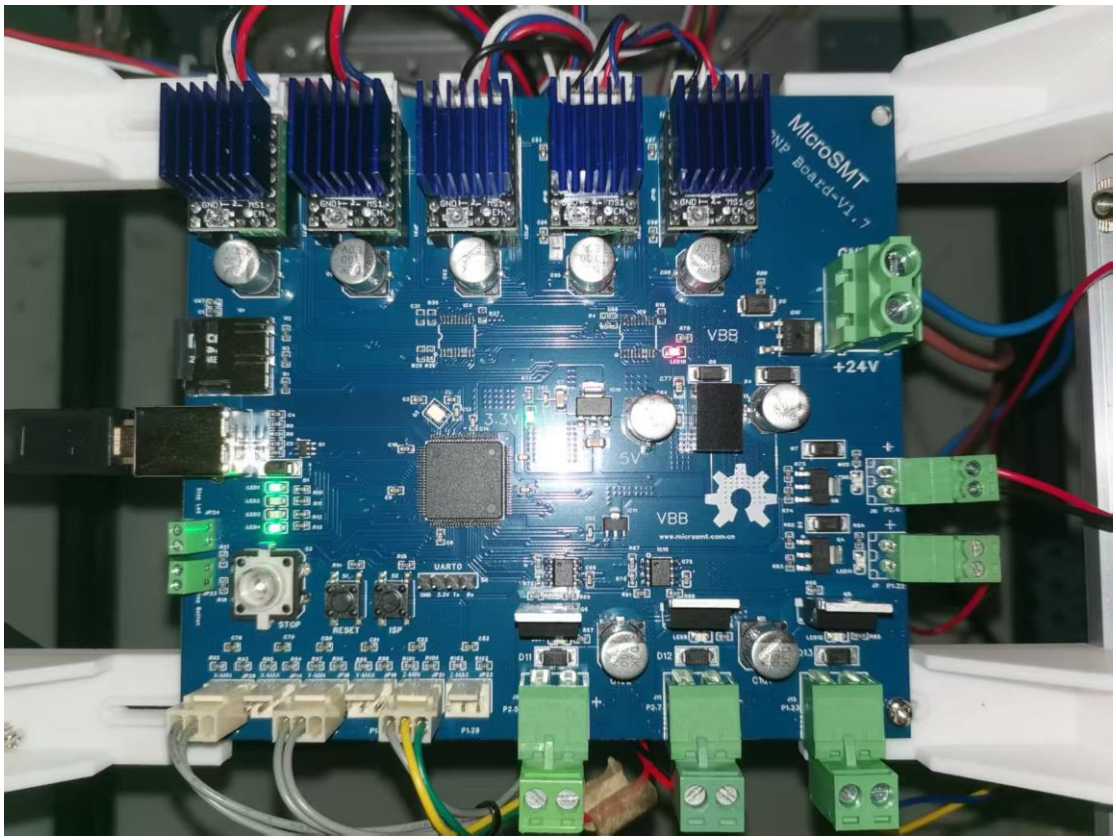
Step 21：Connect air tube.



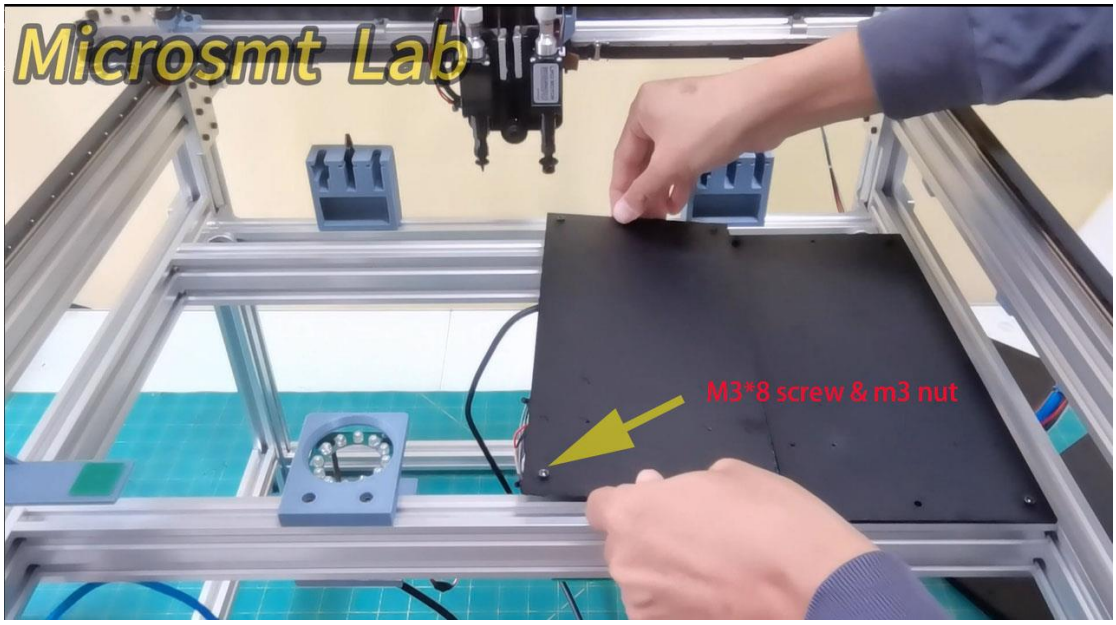
Step 22：Connect all cables to the motherboard.

Tip 1: You need to read the motherboard manual to complete this step.

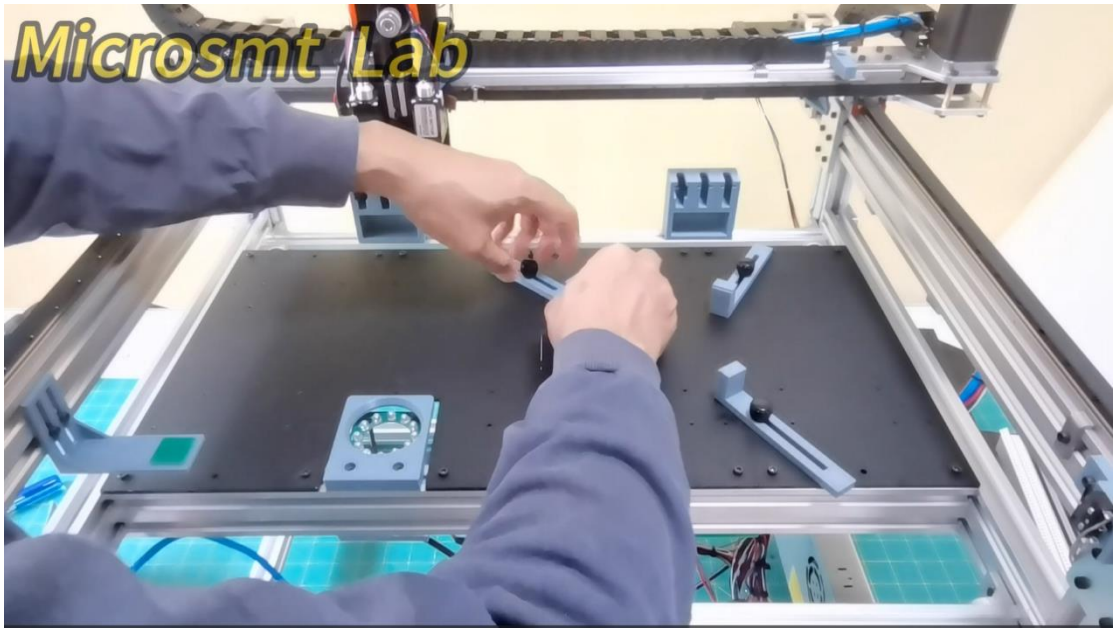
Tip 2: 24VDC has reverse connection protection, you don't have to worry about reverse connection.



Step 23：Place aluminum alloy platform.



Step 24：Install PCB bracket.



Complete the assembly of the electrical . Please refer to the video to learn how to debug PNP.